Configuring the Logger unit

Prior to use, the logger unit will require configuration. Configuration requires that the ESP32 board is connected to a USB of a computer and that the computer has a Serial terminal application installed. The serial terminal can be of any type such as Visual Studio Code or YAT.

The serial terminal should be configured as 115200 baud, No parity, 8 bit, 1 stop bit and no flow control. Once configured and connected, open the serial terminal. This will display text as the ESP32 boots up. Once the boot process is complete then press ? on the terminal. The list of configuration commands will be displayed.

?	- Display help
A	- Show SSID and Keyr
A SSID Key	- Set SSID and Key
H	- Display Hostname
H Hostname	- Set Hostname
I Yes	- Re-initialise Flash memory
Df	- Display flash data
Ds	- Display system data
Dr	- Display RFM69 registers
R Yes	- Restart the ESP
Z	- Set the parameters for your time zone

The first parameter is the NTP server address
The second parameter is your time zone's offset in seconds from UTC
The third parameter is the DST offset in seconds

Example (UK) Z pool.ntp.org 0 3600
Example (Singapore) Z pool.ntp.org 28800 0
Example (Barbados) Z pool.ntp.org -14400 0
Example (Alaska) Z pool.ntp.org -32400 3600

In order to connect to your wireless network, you will need the SSID and Key for your modem or wireless access point.

Find the SSID and Key and enter:

A SSID KEY where "SSID" and "KEY" are the SSID and key of your wireless network. An example might be:

A PLUSNET-ABCD 7abcdef6a4

Now enter

R Yes (the Yes is case sensitive)

This will cause the ESP32 to reboot. Watch the bootup information and provided the SSID and key have been correctly entered you should see the ESP32 connect to your network and collect the current time. The only other configuration required is your local time zone. This configuration sets the NTP server, your offset from UTC (GMT) and your DST offset during the period when DST is in operation, both are in seconds. In the UK the offset from UTC (GMT) is zero and the daylight-saving time offset is 3600. The DST offset is regarded as summer time offset relative to Winter time.

To enter the time zone information for the UK (UTC-0) would be:

Z pool.ntp.org 0 3600

This would set no offset from UTC and one-hour positive offset for DST when it is operating.

To enter the time zone information if you are in New York, USA (UTC-5)

Z pool.ntp.org -18000 3600

To enter the time zone information if you are in Sydney, Australia (UTC+10)

Z pool.ntp.org 36000 3600

To enter the time zone information if you are in Togo, West Africa which does not observe DST

Z pool.ntp.org 0 0

Set the hostname of your logger. The default hostname is WS_LOGGER you may change this if required

H New Hostname

example

H WS_LOGGER_2

This completes the configuration.